$$I_L = \frac{E_0}{\omega L}\sin\left(\omega t - \frac{\pi}{2}\right)$$

Here ωL is the reactance of the inductance.

The current through the capacitor leads the applied emf by $\pi/2$ and is given by

$$I_C = \frac{E_o}{(1/\omega C)}\sin\left(\omega t + \frac{\pi}{2}\right)$$

Here, $(1/\omega C)$ is the reactance of the capacitance.

The total current through the circuit is given by

$$I = I_L + I_C$$

$$= \frac{E_0}{\omega L}\sin\left(\omega t - \frac{\pi}{2}\right) + \frac{E_0}{1/\omega C}\sin\left(\omega t + \frac{\pi}{2}\right)$$

$$= -\frac{E_0}{\omega L}\cos\omega t + \frac{E_0}{(1/\omega C)}\cos\omega t$$

or $$I = E_0\left(\omega C - \frac{1}{\omega L}\right)\cos\omega t \qquad ...(1)$$

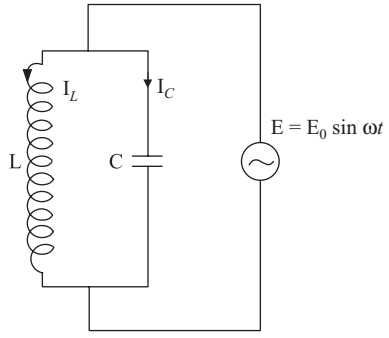


FIG. 13.14

If for a particular frequency v_0, $\omega C = \frac{1}{\omega L}$, then from Eq. (1), the current becomes zero.

The circuit now offers an infinite impedance. Such a circuit which offers an infinite impedance to the A.C. is called a parallel resonant circuit. The frequency is called the resonant frequency.

Thus at v_0, we have,

$$\omega C = \frac{1}{\omega L}$$

or $$\omega = \frac{1}{\sqrt{LC}}$$

or $$2\pi v_0 = \frac{1}{\sqrt{LC}}$$

$\therefore$ $$v_0 = \frac{1}{2\pi\sqrt{LC}} \qquad ...(2)$$

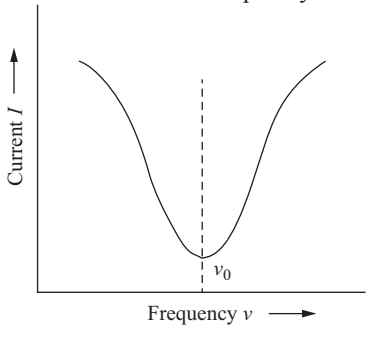


FIG. 13.15

The variation of the current with frequency is shown in Fig. 13.15.

13.3 AC CIRCUIT CONTAINING RESISTANCE, INDUCTANCE AND CAPACITANCE IN SERIES (SERIES RESONANCE CIRCUIT)

Let an alternating emf $E = E_0 \sin \omega t$ be applied to a circuit containing a resistance R, inductance L and capacitance C in series (Fig. 13.16). Let at any instant, I be the current in the circuit and Q be the charge on the capacitor.

The potential drop across the resistance = RI

The E.M.F. induced in the inductance = $L\frac{dI}{dt}$

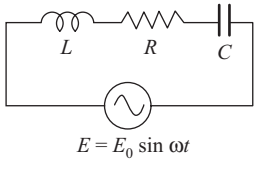


FIG. 13.16

The potential across the plates of the capacitor = Q/C.

$$\therefore \quad L\frac{dI}{dt} + RI + \frac{Q}{C} = E_0 \sin \omega t.$$

Differentiating with respect to t,

$$L\frac{d^2I}{dt^2} + R\frac{dI}{dt} + \frac{1}{C}\frac{dQ}{dt} = E_0\,\omega \cos \omega t \qquad \ldots(1)$$

Let the trial solution be of the form

$$I = I_0 \sin (\omega t - \phi) \qquad \ldots(2)$$

where I_0 and ϕ are constants to be determined.

$$\frac{dI}{dt} = I_0 \omega \cos (\omega t - \phi)$$

and

$$\frac{d^2I}{dt^2} = -I_0\omega^2 \sin (\omega t - \phi)$$

Substituting these value of I, $\frac{dI}{dt}$ and $\frac{d^2I}{dt^2}$, in Eq. (1), we get

$$-LI_0\,\omega^2 \sin (\omega t - \phi) + RI_0\omega \cos (\omega t - \phi) + \frac{I_0}{C}\sin (\omega t - \phi) = E_0\,\omega \cos \omega t$$

or

$$\left(-L\omega^2 + \frac{1}{C}\right) I_0 \sin (\omega t - \phi) + R\omega I_0 \cos (\omega t - \phi)$$

$$= E_0\omega\,[\cos (\omega t - \phi) + \phi]$$

$$= E_0\omega\,[\cos (\omega t - \phi)\cos\phi - \sin(\omega t - \phi)\sin\phi]$$

Equating the coefficients of $\sin (\omega t - \phi)$ and $\cos (\omega t - \phi)$ on either side,

$$\left(-L\omega^2 + \frac{1}{C}\right) I_0 = -E_0\omega \sin \phi, \qquad \ldots(3)$$

and

$$R\omega I_0 = E_0\omega \cos \phi. \qquad \ldots(4)$$

Dividing Eq. (3) by Eq. (4), we get

$$\tan \phi = -\frac{\left(-L\omega^2 + \frac{1}{C}\right)}{R\omega} = \frac{\omega L - \frac{1}{C\omega}}{R} \qquad \ldots(5)$$

Squaring and adding Eqs. (3) and (4), we get

$$I_0^2\left[\left(-L\omega^2 + \frac{1}{C}\right)^2 + R^2\omega^2\right] = E_0^2\omega^2$$

or

$$I_0^2\left[R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2\right] = E_0^2$$

or

$$I_0 = \frac{E_0}{\sqrt{\left[R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2\right]}} \qquad \ldots(6)$$

Substituting the value of I_0 in Eq. (2), we get

$$I = \frac{E_0}{\sqrt{\left[R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2\right]}} \sin(\omega t - \phi) \quad \ldots(7)$$

where $\phi = \tan^{-1} \dfrac{\omega L - \dfrac{1}{\omega C}}{R}$

Eq. (7) represents the current at any instant.

The quantity $\sqrt{\left[R^2 + \left(\omega L - \dfrac{1}{\omega C}\right)^2\right]}$ is the *impedance Z* of the circuit.

$L\omega$ and $1/\omega C$ respectively represent inductive reactance X_L and capacitive reactance X_C. Thus $Z = \sqrt{[R^2 + (X_L - X_C)^2]}$.

The current lags in phase behind e.m.f. by an angle

$$\phi = \tan^{-1} \frac{\omega L - \dfrac{1}{\omega C}}{R} = \tan^{-1} \frac{X_L - X_C}{R}$$

The following three cases arise:

(*i*) When $X_L > X_C$, ϕ is positive so that the *current lags behind the applied emf.*

(*ii*) When $X_L < X_C$, ϕ is negative, so that the *current leads the applied emf.*

(*iii*) When $X_L = X_c$, $\phi = 0$, and the *current is in phase with the emf.*

Series Resonant Circuit

The value of current at any instant in a series *LCR* circuit is given by

$$I = \frac{E_0}{\sqrt{\left\{R^2 + \left(\omega L - \frac{1}{C\omega}\right)^2\right\}}} \sin(\omega t - \phi) \quad \ldots(1)$$

where $\sqrt{\left\{R^2 + \left(\omega L - \dfrac{1}{C\omega}\right)^2\right\}} = Z$

is called the impedance of the circuit.

At a particular frequency, $\omega L = \dfrac{1}{\omega C}$ so that the impedance becomes minimum being given by $Z = R$. *This particular frequency v_0 at which the impedance of the circuit becomes minimum and, therefore the current becomes maximum, is called the* ***resonant frequency*** *of the circuit. Such a circuit which admits maximum current is called series resonant circuit.*

Thus at v_0, we have

$$\omega L = \frac{1}{\omega C} \quad \text{or} \quad 2\pi v_0 L = \frac{1}{2\pi v_0 C}$$

or

$$v_0 = \frac{1}{2\pi\sqrt{LC}}$$

The maximum current in the circuit = $I_0 = E_0/R$. The variation of current with frequency of applied voltage is shown in Fig. 13.17. The sharpness of peak depends upon the resistance R of the circuit. For low resistance, the peak is sharp.

Acceptor Circuit. The series resonant circuit is often called an 'acceptor' circuit. By offering *minimum impedance* to currents at the resonant frequency, it is able to select or accept most readily the current of this one frequency from among those of many frequencies.

In radio receivers, the resonant frequency of the circuit is tuned (by changing C) to the frequency of the signal desired to be detected.

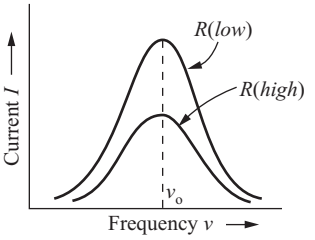


FIG. 13.17

Voltage Magnification. At resonance, the (peak) current through the circuit is

$$I_0 = \frac{E_0}{R}$$

The (peak) voltage across the inductance is

$$V_L = X_L\, I_0 = \omega L \times \frac{E_0}{R} = \frac{\omega L}{R}(E_0) = Q\, E_0,$$

where Q is known as the *quality factor* of the circuit. Thus at resonance the voltage drop across L (which is also the voltage across C) is Q times the applied voltage. Hence the chief characteristic of the series resonant circuit is 'voltage magnification'. This voltage magnification does not increase the power in the circuit as the reactive component of the power is *wattless*.

The Q-factor

$$\text{Q-factor} = \frac{\text{Reactance of the coil at resonance}}{\text{Resistance of the circuit}} = \frac{L\omega_0}{R}$$

Q-factor determines the degree of selectivity of the circuit while tuning. This is because, for larger values of Q-factor the frequency response curve of the circuit is a steep narrow peak. For smaller values of Q-factor, the frequency response curve is quite flat (Fig. 13.18).

Sharpness of resonance of a series L-C-R circuit.

The sharpness of resonance is defined as the rapidity with which the current falls from its resonant value (E_0/R) *with change in applied freqeuency*.

A curve with low value of R falls very sharply on either side than that with high value of R (Fig. 13.18). Resonance in the first case is said to be *sharper* than in the second.

Vector Diagram. A vector diagram of a series LCR circuit is shown in Fig. 13.19. Since L-C-R are connected in series, the current through each is same. Let E_R. E_L and E_C be the potential drops across resistance, inductance and capacitance.

The vector $E_R = I_0R$ is *in phase with* the current.

The vector $E_L = \omega LI_0$ is 90° *advance of* the current.

The vector $E_C = I_0/\omega C$, is 90° *behind* the current.

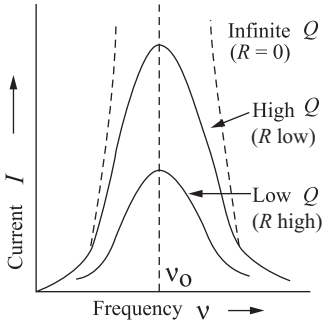


FIG. 13.18

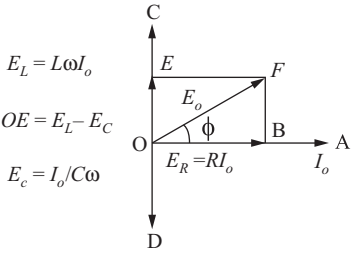


FIG. 13.19